distribution of the nutrients, but may lead to undesirable loss of nutrients and flavor changes. Thus, some cereals are exposed to multiple coating processes for topical application of the added nutrients. A phase 1 coating may include vitamin addition, then phase 2 may include coating with slurries of sugars, honey and flavoring agents. Coatings may be applied by spraying the product as it passes down a conveyer belt or may be added by means of a coating drum (Burns, 2000). Manufacturers generally add nutrients at a higher level than labeled to compensate for possible losses during processing, thus ensuring that content of fortification nutrients in the packaged cereal meets or exceeds the declared level (FDA, 2010).

Nutrient data: Due to the frequency of reformulations of breakfast cereals and brand name specificity of most items in this food group, the Nutrient Data Laboratory relies heavily on the cereal industry to provide current nutrient data for breakfast cereals in SR. Kellogg and General Mills, who represent nearly two-thirds of the RTE retail market (Schroeder, 2011), typically supply data each year, while Quaker, Post, and others contribute data some years. Breakfast cereal manufacturers generally can provide data for proximates, all fortification vitamins and minerals and some non-fortification vitamins and minerals. Data for fatty acid classes (total saturated, monounsaturated and polyunsaturated fatty acids) are generally provided, but individual fatty acids rarely so. Industry-provided fortification nutrient values are based on the label-declared values, representing the minimum amount of that fortified nutrient that should be present in the cereal. Although industry does not provide values for all non-fortification vitamins and minerals, a portion of these nutrients (e.g., magnesium and vitamin C) are generally industry-supplied. Some nutrient values are derived from the product's nutrition facts label, as well.

Every few years, beginning in 2002, various RTE cereals with a high market share have been selected for statistically representative nationwide sampling and nutrient analysis as part of the USDA National Food and Nutrient Analysis Program (NFNAP). Hot cereals, such as regular and instant oatmeal, corn grits, and farina, were sampled through NFNAP, as well. The NFNAP sampling method is described in detail elsewhere (p. 52). The most recent sampling in 2013 was for Quaker and private label maple brown sugar instant oatmeal, General Mills Honey Nut Cheerios and Cinnamon Toast Crunch.

Approximately 200 breakfast cereals are included in a subset of foods supplied for the Food and Nutrient Database for Dietary Studies, which is used for national nutrition monitoring. For these products, there is a list of 65 nutrients for which values must be provided. A variety of standard imputing methods are available in NDL's databank system for estimating missing nutrient values. The predominant imputation method for RTE cereals is by NDL's formulation estimation procedures. These estimation procedures were incorporated into the databank system; they include linear programming techniques to estimate ingredient proportions by weight and calculate a full nutrient profile based on this estimated commercial recipe (i.e. formulation) (Haytowitz, 2009). Individual fatty acids, choline, vitamin K, carotenoids, caffeine and theobromine are generally derived by the formulation method. In the absence of

analytical data, added folic acid is calculated by subtracting estimated natural food folate from the total folate value provided by the manufacturer.

In general, a profile is calculated by recipe for the cooked version of hot cereals that are sold in bulk, such as rolled oats or farina. The yield and retention factors are applied to the recipe to estimate the effects of cooking on moisture and nutrient levels.

Food Group 08 items in SR include data for both ready-to-eat and hot breakfast cereals that are derived from cereal manufacturers, food labels, lab analyses, formulation and other estimations. Recent trends show a decrease in sugar and sodium levels and increase in fiber levels in RTE cereals, on average (Thomas, 2013). NDL will continue to monitor these and other changes.

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## Cereal Grains and Pasta (Food Group 20)

There are over 180 food items in the Cereal Grains and Pasta food group in the SR. The sources of nutrient data for these items are mainly analytical obtained from the scientific literature or analytical studies. These include data collected by nationwide sampling under NDL's NFNAP program, described earlier.

Federal Definitions and Standards of Identity have been published for a number of cereal grain and pasta products appearing on the market today (FDA, 2008a, 2008b). Federal Enrichment Standards exist for wheat flour, cornmeal, rice, and macaroni and noodle products (FDA, 2008a, 2008b). These standards do not mandate the enrichment of the products, but if it is labeled as "enriched," specified nutrient levels must be present. Federal standards specify enrichment levels or ranges for thiamin, riboflavin, niacin, iron, and folic acid in most enriched products. The Federal Enrichment Standard for riboflavin in enriched rice has been stayed since 1958, and hence riboflavin is not currently added to enriched rice. The standards for enriched grain products were amended to require the addition of folic acid beginning in 1998. Addition of calcium to most enriched products is optional, but if added must meet specified levels. However, consistency in levels of enrichment may be an issue (Guerrero et al, 2009). The current Federal Enrichment Standards are listed in Table 21.

In the Cereal Grains and Pasta food group, data are presented for the enriched and unenriched forms of commonly enriched products.

Table 21. Standards for Enrichment[1]

| Food Item | Thiamin | Riboflavin | Niacin | Iron | Folic Acid | Calcium[2] |
|---|---|---|---|---|---|---|
| | --- milligrams per pound --- | | | | | |
| Wheat flour | 2.9 | 1.8 | 24 | 20 | 0.7 | 960 |
| Self-rising wheat flour | 2.9 | 1.8 | 24 | 20 | 0.7 | 960 |
| Cornmeal | 2.0-3.0 | 1.2-1.8 | 16-24 | 13-26 | 0.7-1.0 | 500-750 |
| Self-rising cornmeal | 2.0-3.0 | 1.2-1.8 | 16-24 | 13-26 | 0.7-1.0 | 500-750 |
| Rice | 2.0-4.0 | 1.2-2.4[3] | 16-32 | 13-16.5 | 0.7-1.4 | 500-1,000 |
| Macaroni and noodle products | 4.0-5.0 | 1.7-2.2 | 27-34 | 13-16.5 | 0.9-1.2 | 500-625 |

1. A range of figures indicates minimum and maximum levels. A single figure is the minimum level, with overages left to good manufacturing practice.
2. Calcium enrichment is optional in these products.
3. The enrichment standard for riboflavin in enriched rice has been stayed since 1958.

In 2010, the US Departments of Agriculture and Health and Human Services released the seventh edition of the Dietary Guidelines for Americans, which set evidence-based

recommendations for the public to help prevent disease (US Departments of Agriculture and Health and Human Services, 2010). The guidelines emphasize consumption of whole grain foods, stating at least half of an individual's recommended total grain intake should be whole grains to reduce the risk of several chronic diseases and help with weight maintenance.

The AACC International definition of whole grains is "Whole grains shall consist of the intact, ground, cracked or flaked caryopsis, whose principal anatomical components - the starchy endosperm, germ and bran - are present in the same relative proportions as they exist in the intact caryopsis."

In 2006, FDA issued draft guidance for Industry and FDA Staff on Whole Grain Label Statements. It can be accessed at: http://www.fda.gov/Food/GuidanceComplianceRegulatory Information/GuidanceDocuments/FoodLabelingNutrition/ucm059088.htm.

Cereal grains.—The majority of the cereal grains included in Food Group 20 are cultivated grasses belonging to the Poaceae (alt.Gramineae) family and are thus true cereals. Amaranth, buckwheat, and quinoa differ botanically from true cereals, and are referred to as pseudo cereals because they are grown and used like cereal grains (Brouk, 1975). Arrowroot flour is derived from arrowroot and tapioca is produced from cassava root, which are both non-cereal-grain plants, but used in ways similar to cereal grains.

The scientific name is given for the most unprocessed form of the cereal grain in the database. The Germplasm Resources Information Network (GRIN) was used as the basic reference for the scientific names and preferred common names (USDA, 2011).

With the exception of corn (maize), which is native to the Americas, nearly all true cereal grains originated in Europe and Asia (Brouk, 1975). Buckwheat is native to central Asia. Amaranth and quinoa are native to Central and South America, respectively.

Kasha, a buckwheat product, originated in Russia. Buckwheat groats, which are roasted to develop a distinctive nutty flavor, may be packaged in the whole form or milled to either coarse, medium, or fine granulations. Kasha is usually cooked as a hot cereal or prepared in combination with other foods and ingredients.

Corn and corn products appearing in Food Group 20 are restricted to field corn varieties and do not represent the varieties (sweet corn) used mainly as a vegetable. Corn and cornmeal products are available in white, yellow, and blue varieties. Yellow corn varieties have higher vitamin A values due to the presence of the provitamin-A carotenoids, alpha- and beta-carotene. Yellow corn also has much higher levels of lutein + zeaxanthin. With the exception of these nutrients, the composition profiles of white and yellow corn are similar.

Self-rising cornmeals and wheat flours have more calcium, phosphorus, and sodium due to the addition of chemical leavening agents and salt. Sodium bicarbonate, monocalcium phosphate, sodium acid pyrophosphate, and sodium aluminum phosphate are the most commonly used leavening agents. Salt is also usually added to self-rising products for flavor. Bolted cornmeal has had most of the bran removed during milling, but contains most of the germ present in the whole-grain corn.

Masa corn flour is milled from corn which has been steeped in a lime (calcium hydroxide) solution. This is done both to facilitate the removal of the outer hull of the corn grain and to impart the characteristic flavor of authentic corn tortillas and other related products. As a result of the use of lime in processing, masa corn flour is higher in calcium than other corn products.

Brown rice has the bran layers intact. Rice that has been milled to remove the bran layers is referred to as white rice in this database.

Bulgur, a wheat product, has been produced in the Middle East and northern Africa since ancient times. Bulgur is produced by parboiling, drying, and then cracking wheat kernels. It is usually consumed as a cooked cereal or as an ingredient in other dishes.

Couscous is coarse-ground wheat endosperm made from durum wheat or another hard wheat variety. Couscous is a popular food in northern Africa and in the Middle East. It is usually eaten as a hot cereal or combined with other foods.

Wheat flour tortilla mix is used for making flour tortillas and other related products. This product is higher in calcium than other wheat flour products because calcium carbonate is added.

Bread flour, approximately 13% protein, is milled primarily from hard wheats. Cake flour, approximately 9% protein, is milled from soft wheats. Semolina is coarse-ground endosperm from durum wheat, and is used chiefly for making pasta.

Teff is an ancient crop believed to have been domesticated in the northern highlands of Ethiopia. It is used alone or in combination with sorghum to prepare the fermented flat bread, injera (Dendy, 1995).

Corn grits, farina, rolled oats or oatmeal, and toasted wheat germ are included in Food Group 08, Breakfast Cereals.

Nutrient data for different forms and products of each cereal grain were not obtained from the same sample or source. For example, a single source of wheat was not processed to all forms given in the database: whole-grain, bran, germ, and various flour products. The data were obtained from many sources at different times for analysis and are affected by different variables: growing locations, crop years, cultivars, natural variability, milling and processing techniques, laboratories, and possibly methods of analysis. Therefore, in a comparison of different forms and products of a cereal grain,

nutritional differences may not measure precisely the effect of processing or preparation methods.

Pasta.—Under Federal Standards of Identity, there are two broad categories of pasta products: macaroni and noodle products (FDA, 2008b). Macaroni products are formed by extrusion of the pasta dough into a variety of shapes and sizes including elbows, spirals, shells, twists, wheels, etc. Specific shapes of macaroni products have unique names such as rigatoni, manicotti, ziti, linguini, and spaghetti which are recognized by the consumer.

Although spaghetti is defined under Federal standards as a macaroni product, it is included as a separate category due to its unique market identity. However, the nutrient composition of spaghetti and that of other forms of macaroni products are the same on an equal weight basis.

Noodle products are also available in a variety of sizes and shapes. Federal Standards of Identity specify that noodle products must contain not less than 5.5 percent by weight of the solids of egg or egg yolk (FDA, 2008b).

Various forms of vegetable macaroni and noodle products are available today. Federal standards specify that these products must contain a minimum of 3 percent by weight of the solids of tomatoes (red varieties), artichoke, beet, carrot, parsley, or spinach ((FDA, 2008b). Spinach noodles and tricolor-type (red, green, and regular) macaroni are the most commonly available products of this type on the market.

Protein-fortified macaroni products, both with and without added vegetable solids, are also available. These products usually contain wheat germ, dried yeast, or other ingredients which increase the protein content of the product. If a macaroni product is labeled as “with Fortified Protein,” under Federal standards it must have a protein content of at least 20 percent on a 13-percent moisture basis and protein quality not less than 95 percent of that of casein (FDA, 2008b).

Corn pasta is available on the market to meet the needs of those who are allergic to wheat and hence must avoid foods containing wheat ingredients. Corn pasta is made exclusively from corn flour. Since it contains no wheat flour ingredients, corn pasta is not required to meet Federal standards for macaroni or noodle products.

Fresh-refrigerated pasta has a higher moisture content than dry pasta and must be kept under refrigeration until prepared. Data are presented for plain and spinach types, both of which contain egg. Stuffed pasta such as ravioli and tortellini are listed in Food Group 22, Meals, Entrees, and Side Dishes.

Data are presented for the cooked forms of both egg-containing and non-egg-containing homemade pasta. The recipe used for each item is footnoted.

Oriental noodles do not fall under Federal Standards of Identity. Although these products may be labeled as noodles, they usually do not contain eggs. Chinese-style pasta products currently in SR include rice noodles, chow mein noodles, and fried flat noodles. Two Japanese noodles are currently in SR: soba noodles are made with buckwheat flour; somen is a thin wheat flour noodle. Chinese cellophane noodles, also called long rice noodles, are made from mung bean flour and are included in Food Group 16, Legumes and Legume Products.

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## Eggs (Food Group 01)

Recently, NDL arranged to have regular large, whole eggs picked up in a nationwide sampling as part of NFNAP. Notes on other food items in this food group will be included at a later time.

**Sampling and analysis.** Whole egg samples of regular large eggs were picked up in March/April 2010 at the 12 NFNAP sampling locations. The sample units were sent to the Food Analysis Laboratory Control Center (FALCC) at Virginia Tech for preparation of analytical samples to be sent to the qualified analytical laboratories. Individual samples from each of the 12 locations were prepared for the determination of proximates (moisture, protein and fat), fatty acids, and cholesterol. Samples units from the 12 locations were paired, using randomization, to create six city-pair analytical composites for analysis of vitamins, minerals, and sugars. FALCC also sent quality control (QC) samples to the analytical laboratories to monitor accuracy and precision of measurements.

**Results**. The analytical and QC data received from the analytical laboratories were reviewed. The QC results were found to be acceptable, and the analytical data for most nutrients were comparable to the current data in the National Nutrient Database for Standard Reference, Release 22 (SR22). Values for cholesterol, vitamin D, and vitamin $B_{12}$ were significantly different from the older values which were based on analyses from eggs sampled in 2002.

Cholesterol was determined by gas chromatography; the new value for cholesterol is 372 mg/100g compared to the SR22 value of 423 mg/100g. This estimate was based on the analysis of the 12 individual samples by each of three independent qualified laboratories. The results for QC materials from all three laboratories were also acceptable.

Vitamin D was determined by HPLC-UV for the six city-pair composites. The results for the QC material were within the acceptable range. Two of the city-pair composites were analyzed by another lab using HPLC-MS/MS to double check the data. The inter-lab results were in good agreement.

Values for four of the six city-pair composites for Vitamin D averaged 1.2 µg (49.2 IU)/100g (with a range of 1 µg (39 IU) – 1.8 µg (71 IU)/100g), compared to the SR22 value of 1.2 µg (49.6 IU)/100g. However, the values for the other two city-pairs were 3.8 µg (150 IU)/100g and 8.7 µg (348 IU)/100g. Each city-pair with a higher vitamin D value contained samples of eggs for a specific brand picked up in two non-contiguous states from the same grocery store chain, and those cartons were labeled as follows: 5X MORE VITAMIN D PER EGG.

Reserve samples for the four individual cities that were part of the two city-pairs with higher vitamin D were sent to the laboratory to be analyzed for vitamin D. One of the city values was within the expected range (1 µg (39 IU) – 1.8 µg (71 IU)/100g) of the

data for the current estimate. The other three city values were much higher and fell between 7.1 µg (284 IU)/100g and 12.1 µg (483 IU)/100g. Two of these three samples were from the same store brand, and their respective cartons were labeled as indicated above, and thus higher in vitamin D. The third of the three was obtained from a store brand that had no vitamin D statement on the carton. Additional samples of that brand were sampled, analyzed, and confirmed.

To calculate the final estimate of vitamin D in large, whole eggs, all values from samples which had no vitamin D claims were averaged together to yield a value of 2.0 µg (82 IU)/100g with a range of 1.0 µg (39 IU)/100g to 9.2 µg (368 IU)/100g. The new value is 64% higher than the SR22 value of 1.2 µg (50 IU)/100g.

The values for the store brand which contained a vitamin D claim were not used. NDL staff decided that the presence of a claim could influence the selection of that brand by the consumer and may bias the representativeness of the sample set. However, it is clear that some eggs in the marketplace now contain higher levels of vitamin D. It is likely that this change is due to the fortification of specific feeds given to the laying hens. More research will be needed to assess the impact on vitamin D levels in eggs nationwide.

The new value for vitamin $B_{12}$ (0.89 µg/100g) is 31% lower than the value in SR22 (1.29 µg/100g). The values for the QC samples were satisfactory.

**Impact**. All egg products that contain egg yolk, where the fat soluble cholesterol and vitamin D are found, were updated to reflect the change in values. NDL food specialists, who use whole eggs and other egg products as ingredients in formulations and recipes, will use these cholesterol and vitamin D values to calculate the composition of those food items. NDL plans to follow up on the sampling and analysis of whole eggs in one to two years to monitor levels of vitamin D in samples nationwide.